THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH2014: Multivariable Calculus and Linear Algebra Part I
Dec 17, 2020 TIME: 2:30 p.m.-4:30 p.m. (include both Part I and Part II)

1. (4 marks)

Compute the value of $\int_0^2 \int_0^2 (x+y)^{100} dx dy$.

Select one:

- a. $\frac{4^{102}-2^{103}}{10302}$
- b. $\frac{4^{101}-2^{101}}{10302}$
- c. $\frac{4^{102}-2^{103}}{5151}$
- d. $\frac{4^{102}-2^{103}}{20604}$

2. (4 marks)

Find the area of the region bounded by $y^2 = 2px + p^2 (p > 0)$ and $y^2 = -2qx + q^2 (q > 0)$.

Select one:

- a. $\frac{2}{3}(p+q)\sqrt{pq}$
- b. $\frac{1}{3}(p+q)\sqrt{pq}$
- c. $\frac{1}{3}(p+q)\sqrt{p+q}$
- d. $\frac{2}{3}(pq)\sqrt{p+q}$

3. (4 marks)

Let $0 < a < b$, evaluate $\iint_D \sqrt{x^2+y^2}\,dA$, where $D = \{(x,y) \mid a^2 \leqslant x^2+y^2 \leqslant b^2\}$.

Select one:

- a. $\frac{4}{3}\pi(b^3+a^3)$
- b. $\frac{2}{3}\pi(b^3+a^3)$
- c. $\frac{1}{3}\pi(b^3-a^3)$
- d. $\frac{1}{3}\pi(b^3+a^3)$
- e. $\frac{4}{3}\pi(b^3-a^3)$
- f. $\frac{2}{3}\pi(b^3-a^3)$

4. (4 marks)

Let $a, b, c > 0$, evaluate $\iiint_D z^2 dV$, where D is the region enclosed by the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.

Select one:

- a. $\frac{2\pi abc^3}{15}$
- b. $\frac{8\pi abc^3}{15}$
- c. $\frac{4\pi abc^3}{15}$
- d. $\frac{8\pi abc}{15}$
- e. $\frac{4\pi abc}{15}$
- f. $\frac{2\pi abc}{15}$

5. (4 marks)

Find the centroid of the region that is bounded by $x^2 + y^2 = 2y$ and $x^2 + y^2 = 4y$.

Select one:

- a. $(1, \frac{8}{3})$
- b. $(0, 3)$
- c. $(1, 2)$
- d. $(0, 1)$
- e. $(1, 1)$
- f. $(0, \frac{8}{3})$
- g. $(0, \frac{7}{3})$
- h. $(1, \frac{7}{3})$
- i. $(1, 3)$

6. (3 marks)

Which of the following is a reduced row echelon form of the matrix below?

$$\begin{bmatrix} 1 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 1 & 2 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Select one:

○ a. $\begin{bmatrix} 1 & 0 & 2 & 0 & 3 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

○ b. $\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

○ c. $\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

○ d. $\begin{bmatrix} 2 & 0 & 2 & 0 & 2 \\ 0 & 3 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

7. (3 marks)

How many of the following matrices are in row echelon form? How many are in reduced row echelon form?

$$\begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Select one:

- a. 3; 1
- b. 3; 2
- c. 4; 3
- d. 2; 1
- e. 4; 1

8. (3 marks)

How many free variables are there in the following system of the linear equations

$$\begin{cases} x_1 + 3x_2 - 5x_3 - 2x_4 = 0 \\ -3x_1 - 2x_2 + x_3 + x_4 = 0 \\ -11x_1 - 5x_2 - x_3 + 2x_4 = 0 \\ 5x_1 + x_2 + 3x_3 = 0 \end{cases}$$

Select one:

- a. 3
- b. 2
- c. 1
- d. 4
- e. 0

9. (3 marks)

For a homogeneous system of 5 linear equations with 4 unknowns, which of the following is possible?

Select one or more:

- a. The solution set contains infinitely many solutions.
- b. The solution set is empty.
- c. The solution set contains unique solution.
- d. The solution set contains unique non-trivial solution.

10. (3 marks)

Calculate $det(D)$, where $D = \begin{bmatrix} a & b & c & 1 \\ b & c & a & 2 \\ c & a & b & 2 \\ \frac{2b+c}{3} & \frac{2c+a}{3} & \frac{2a+b}{3} & 2 \end{bmatrix}$.

Select one:

- a. None of the choices are correct.
- b. 1
- c. $a + b + c$
- d. 0
- e. $\dfrac{a+b}{2}$

11. (3 marks)

For any $n \times n$ matrix A, find a correct relationship between $det(-A^T)$ and $det(A)$.

Select one:

- a. $det(-A^T) = det(A)$
- b. $det(-A^T) = (-n)det(A)$
- c. None of the choices are correct.
- d. $det(-A^T) = (-1)^n det(A)$
- e. $det(-A^T) = -det(A)$

12. (3 marks)

For four vectors $\vec{u} = (1, 4, -7), \vec{v} = (2, -1, 4), \vec{w} = (0, -9, 18), \vec{p} = (0, -9, 17)$, check which of the followings is (are) correct:

Select one or more:

- ☐ a. $\vec{u}$ can be represented by linear combinations of $\vec{v}, \vec{w}$
- ☐ b. $\vec{w}, \vec{v}, \vec{p}$ are linearly dependent
- ☐ c. $\vec{u}, \vec{v}, \vec{w}$ can be placed on the same plane
- ☐ d. $\vec{u}, \vec{v}, \vec{p}$ are linearly independent

13. (3 marks)

Which of the following statements is(are) correct:

Select one or more:

- ☐ a. For $n \times n$ matrices A,B and C, if $AC = BC$, then $A = B$.
- ☐ b. If A and B are $n \times n$ non-singular matrices , then $AB = BA$.
- ☐ c. If for any $\vec{x} \in R^3$ we have $\vec{u} \times \vec{x} = \vec{v} \times \vec{x}$, then $\vec{u} = \vec{v}$
- ☐ d. If for one $\vec{x} \in R^3$ we have $\vec{u} \times \vec{x} = \vec{v} \times \vec{x}$, then $\vec{u} = \vec{v}$
- ☐ e. If A and B are $n \times n$ matrices and A is a triangular matrix, then $AB = BA$.

14. (3 marks)

For two matrices $A = \begin{bmatrix} -5 & -4 \\ 8 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & -3 \\ 1 & 0 & -1 \\ 1 & 1 & -2 \end{bmatrix}$, determine whether they are diagonalisable.

Select one:

- a. Only A is diagonalisable.
- b. Both of them are diagonalisable.
- c. None of them are diagonalisable.
- d. Only B is diagonalisable.

15. (3 marks)

Which of the following could possibly be the $(1, 1)$ entry of A^n, where $A = \begin{bmatrix} 6 & -1 \\ 2 & 3 \end{bmatrix}$ and $n \in \mathbb{N}$?

Select one or more:

- a. 2101
- b. 61
- c. 5226
- d. 2020
- e. 994
- f. 369
- g. 186

16. (2 marks)

Let u, v and w be column vectors of the same size. If w is a linear combination of u and v, v is a linear combination of u and w, then u is a linear combination of v and w.

Select one:

- True
- False

17. (2 marks)

Suppose that v_1 and v_2 are two vectors in R^n. Then $span\{v_1, v_2\} = span\{v_1 + v_2, v_1 - v_2\}$.

Select one:

- ○ True
- ○ False

18. (2 marks)

Suppose that v_1 and v_2 are two vectors in R^n. Then $span\{v_1, v_2\} = span\{2v_1, 3v_2\}$.

Select one:

- ○ True
- ○ False

19. (2 marks)

Suppose S is a set that can generate R^3. Then S contains no less than 3 vectors.

Select one:

- ○ True
- ○ False

20. (3 marks)

Which of the following operation will not change the determinant of a matrix?

Select one or more:

- ☐ a. multiply a column by a non-zero constant.
- ☐ b. interchange two rows.
- ☐ c. add a multiple of a column to another column.
- ☐ d. add a multiple of a row to another row.

21. (3 marks)

If A and B are both $n \times n$ matrices with rank k, which of the following must be true?

Select one:

- a. rank$(A + B) \leq k$.
- b. rank$(A + B) = k$.
- c. rank$(A + B) =$rank$(B + A)$.
- d. rank$(A - B) \leq$rank$(A + B)$.

22. (3 marks)

Which of the following must be true if matrix A is **NOT** invertible?

Select one or more:

- [ ] a. The determinant of A is zero.
- [ ] b. The columns of A are linearly independent.
- [ ] c. The rows of A are linearly dependent.
- [ ] d. A row reduces to the Identical matrix.
- [ ] e. A is singular.

23. (3 marks)

A system of linear equations has a unique solution. Which of the following is a possible augmented matrix of the system?

Select one or more:

- ☐ a. $\begin{bmatrix} 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$.
- ☐ b. $\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 2 & 0 & 5 \\ 0 & 0 & 1 & 0 \end{bmatrix}$.
- ☐ c. $\begin{bmatrix} 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.
- ☐ d. $\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$.

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Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL THREE questions.

Notes:

- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- You must start each question on a new page. You should write down the question number on the top right hand corner of each page.
- If your answer to a question spans over more than one page, you must indicate clearly on each page (except the last) that the answer will continue on the subsequent page.
- Some formulae you may need:

For $n \geq 2$:

$$\begin{aligned}
\int \cos^n \theta \, d\theta &= \frac{1}{n}\cos^{n-1}\theta \sin\theta + \frac{n-1}{n}\int \cos^{n-2}\theta \, d\theta \\
\int \sin^n \theta \, d\theta &= -\frac{1}{n}\sin^{n-1}\theta \cos\theta + \frac{n-1}{n}\int \sin^{n-2}\theta \, d\theta \\
(\arctan x)' &= \frac{1}{1+x^2}
\end{aligned}$$

1. (10 marks) Find the absolute maximum and minimum values of $f(x, y, z) = 2x^2 + 3y^2 + 2z^2 - xy + yz - xz$ on the closed ball $x^2 + y^2 + z^2 \leq 1$.

2. (10 marks) Find the mass of a circular cone of base radius r = 1 and height h=4 given that the density equals the distance from the vertex.

   a) Write out the triple integrals by cylindrical coordinates.

   b) Write out the triple integrals by spherical coordinates.

   c) Calculate the result by either a or b.

3. (10 marks) Let

$$A = \begin{bmatrix} 2 & 0 & 2 \\ 3 & 3 & b \\ 2 & 0 & a \end{bmatrix}.$$

   (a) Given that $\lambda = -2$ is an eigenvalue of A and A is diagonalizable, find the values of a and b.

   (b) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

   (c) Find a non-singular matrix P and a diagonal matrix D, such that $A = PDP^{-1}$.

   (d) Find the $(3, 3)$-entry of A^{2020}.

******* **END OF PAPER** *******